Technology Stack

Date	4 july 2026
Team ID	xxxxxx
Project Name	Credit card approval
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** *(e.g., React, Angular, HTML/CSS)*	*HTML&CSS3*	*Used to design the interactive frontend pages (index.html and result.html).* *Provides a clean, structured, and responsive form interface so applicants can easily input their background details and see their prediction status immediately.*

2	**Backend / Server-Side** *(e.g., Node.js, Python/Django, Java)*	*Python (flask framework)*	*Used to write the core routing logic (app.py) that receives user data from the frontend web form.* *It easily integrates with our machine learning libraries to load the model file (model.pkl) and calculate predictions instantly.*
3	**Database / Data Storage** *(e.g., MySQL, MongoDB, PostgreSQL)*	*Csv files/local storage*	*Used to hold the application datasets (credit_record.csv and cleaned_credit_record.csv) for training and data evaluation. Provides a simple, lightweight system that reads data parameters rapidly using standard libraries without needing a complex standalone SQL server.*
4	**Cloud / Hosting / Deployment** *(e.g., AWS, Azure, Heroku, Docker)*	*Github&localost*	*GitHub is chosen for secure cloud version control to save, organize, and back up all code files systematically across phase folders.* *Localhost hosting provides an easy-to-run development environment on our laptops to test the Python web app locally before any production release.*
5	**Version Control & CI/CD**	*Git/github*	** Used for maintaining clear version control and tracking changes across our dynamic machine learning and web source code files.* *Allows seamless backup of project documentation steps directly into assigned trace*

	(e.g., GitHub, GitLab, Jenkins)		
6	**Third-Party APIs / Other Tools** *(e.g., Stripe, Twilio, Google Maps)*	*Scikit-learn*	*Scikit-Learn provides the core analytical libraries required to train the credit risk evaluation model.* *Joblib handles loading the serialized pre-trained parameters (model.pkl and encoders.pkl) to provide optimized, instant data predictions.*